Technical Memo — GRN-Prior Expression Embedding Benchmark

Question. Does a graph-aware encoder using DoRothEA TF->target edges produce *better* pseudobulk expression embeddings than a matched non-graph baseline — where "better" means the embedding captures biological state (cell type, disease), not that it reconstructs expression — and does any benefit hold up under low-data, noise, and graph corruption?

Answer (short). How you use the GRN matters more than whether you use it. - As a **deep-encoder constraint** (hard mask or soft graph-penalty): it *hurts* — consistently below PCA and a plain autoencoder — and a degree-preserving **rewired** graph does as well, so what little it offers is regularization, not the specific biology. - As a **fixed feature transform** (decoupler ULM TF-activity, the classical use): it carries **real** biological signal — it beats **both** a rewired-network TF-activity control *and* a matched-dimension random projection (real 0.847 > rewired-net 0.804 > random 0.775) — and it **beats PCA and the baseline in the low-data regime** (where priors are supposed to help). At full data it merely ties PCA; under heavy noise it fades to the baseline. - The effect is **not DoRothEA-specific** (CollecTRI reproduces and slightly exceeds it) and **replicates on a second dataset** (COVID PBMC). Disease is decodable on the 3-class COVID data (PCA best) but not on the fully donor-confounded 2-class RA data.

A fair, carefully-scoped result: the prior encodes genuine biology, but only helps when applied the classical way and mostly when data is scarce; imposing it on a learned encoder is worse than doing nothing.

Data, task, evaluation

CELLxGENE RA PBMC (`d18736c3…`, 108,717 cells) -> **pseudobulk by (donor x cell_type)**, 536 groups -> **500 kept** (>=10 cells), CP10K+log1p, restricted to the 8,376 genes shared with DoRothEA. Readouts: **cell type** (15 classes, primary) and **disease** (secondary). All scoring is **donor-grouped 5-fold CV**, with folds **re-shuffled per seed** (so each of the 5 seeds is a genuine re-partition, not the same split) and the encoder retrained on train donors *inside* each fold — neither encoder nor probe ever sees a held-out donor. Metric: macro-F1 with a **fixed 15-class label set** (comparable across folds); paired deltas over the 25 (seedxfold) splits.

Is the dataset suitable?

Balanced **18 RA / 18 normal**, **no sex confound** (12F/6M both arms), **single assay** (10x 3' v3 -> no assay–disease confound), 15 cell types. But the design is **between-subjects**, so disease is confounded with **donor identity**. Consequence: cell type is trustworthy; RA disease is reported as suggestive only. (The COVID replication has 3 disease states across 75 donors — still between-subjects, but less degenerate, and there disease *is* decodable.)

Models compared (all share the gene set, capacity where noted, and training budget)

- **PCA** (linear floor) and **baseline** (dense autoencoder) — 64-d, no prior.
- **GRN-as-constraint**: `grn_real` (first layer masked to signed TF regulons), `grn_soft:lambda` (dense layer + penalty shrinking off-regulon weights). Controls at matched density/degree: `grn_rewired`, `grn_sign_shuffled`, `grn_random`.
- **GRN-as-transform**: `dc_tfact` (decoupler ULM TF-activity, 293-d), `dc_tfact_pca` (->PCA-64, dimension-matched), `dc_tfact_collectri` (CollecTRI net, 675-d).
- **Null**: `rand_proj` (random linear features at the TF-activity dimension).

Results (cell-type macro-F1, definitive post-review sweep — `final.csv`, 5 seeds, per-seed folds)

model (dim)	full	lowdata k=8	noise p=0.3
PCA (64)	**0.869**	0.632	0.770
baseline AE (64)	0.785	0.661	0.733
dc_tfact — DoRothEA ULM (293)	0.847	0.686	0.720
dc_tfact_rewired — *rewired net* (293)	0.804	0.661	0.703
rand_proj — *random* (293)	0.775	0.624	0.648
dc_tfact_collectri (675)	**0.869**	**0.746**	**0.784**

model (dim)	full	lowdata k=8	noise p=0.3
dc_tfact_pca (64)	0.783	0.630	0.686
grn_soft:0.001	0.753	0.641	0.715
grn_real (hard mask)	0.714	0.591	0.615
grn_rewired / grn_random (full)	0.665 / 0.754	–	–

Reading it: 1. **GRN-as-constraint hurts.** `grn_real`/`grn_soft` trail both PCA and the baseline everywhere; *more* prior is *worse* (soft 0.001 0.75 > soft 0.01 0.62 > hard 0.71). The problem isn't mask hardness — it's imposing the graph on the encoder at all. 2. **The corruption test kills the "it's biology" story for the constraint.** At full data `grn_random` (0.754) actually tops `grn_real` (0.714) and `grn_rewired` (0.665) — corrupted graphs do as well or better, so any constraint benefit is sparsity/regularization. 3. **The transform is different — and now doubly-controlled.** `dc_tfact` (0.847) beats **both** the rewired-DoRothEA-net TF-activity `dc_tfact_rewired` (0.804) *and* the random projection `rand_proj` (0.775). Real regulons > rewired regulons > random => the signal is the *specific* regulatory structure, not just sparsity or dimensionality. (This rewired-net null was the control the earlier draft lacked.) 4. **It helps most when data is scarce.** At full data `dc_tfact` ~ PCA; under **low data** TF-activity clearly beats baseline and PCA (CollecTRI beats the baseline in **96–100 %** of folds; see `final_stats.csv`), and there `dc_tfact` > `dc_tfact_rewired` — biology, not structure. Under heavy noise `dc_tfact` fades below the baseline while CollecTRI holds. 5. **Not prior-specific.** CollecTRI >= DoRothEA in every condition -> the effect is a property of "TF-activity as a transform," and a broader/better network helps more.

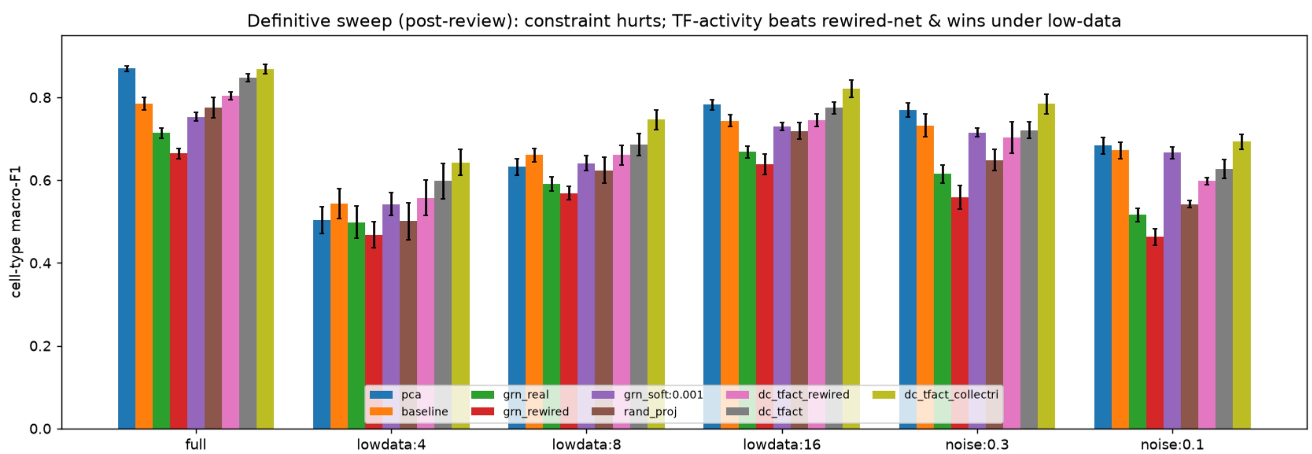

Figure 1: headline

Disease. On RA (2-class, fully donor-confounded) nothing decodes disease from held-out donors (~ chance). On **COVID (3-class, 75 donors)** disease *is* decodable — PCA best (0.715), then TF-activity (~0.66–0.67); the GRN-mask models beat the *overfit* dense baseline (grn_real 0.654 vs baseline 0.588) but `grn_rewired` (0.646) matches them -> again regularization, not biology.

Biology vs batch. Donor-leakage (kNN donor accuracy, lower better): baseline 0.086 < random 0.098 < grn_real 0.114 < … < PCA 0.166. The prior does not reduce donor signal.

External validity. The full ordering (PCA >= TF-activity > baseline ~ GRN-mask > soft-prior) replicates on COVID -> not an artifact of the RA dataset.

Bottleneck-dimension sensitivity. Sweeping z in {32, 64, 128}, the ordering PCA >= baseline ~ soft-prior > hard-mask holds in every condition, and `grn_real` is *always* worst. A wider bottleneck helps the hard mask (full: 0.67 -> 0.71 -> 0.75) but it never reaches the baseline (~0.79) — so the negative-for-constraints result is not a z=64 tuning artifact.

Where to place the prior — encoder vs decoder vs symmetric

The masked encoder infers TF activity from genes (the *inverse* direction); the **decoder** (TF->gene) is the *causal/generative* direction and is where biologically-informed autoencoders (expiMap) put the mask. Testing all three (full-data cell type): **grn_decoder 0.738 > grn_real (encoder) 0.704 > grn_symmetric 0.675** (masking both over-constrains). In *every* placement the real graph beats its rewired control (decoder 0.738 vs 0.715; encoder 0.704 vs 0.661) — a genuine, if modest, biology signal. But **none beats the dense baseline (0.785) or PCA (0.869)**. So a better-placed prior helps *relative to other constrained models*, yet still doesn't beat doing nothing — reinforcing the main conclusion. (`13_decoder_prior.py`, `decoder.csv`.)

Supervised probe vs unsupervised clustering — the metric flips the winner

The probe measures *linear separability*; a label-free **clustering** metric (KMeans, ARI/NMI vs cell type — the scIB-style test) measures *intrinsic structure*. They disagree sharply (`14_ clustering.py`, `clustering.csv`): by ARI, **PCA is worst (0.12)** while GRN-informed **CollecTRI TF-activity is best (0.30)**, with the autoencoders (0.22–0.26) also above PCA. PCA maximises variance/separability but leaves elongated, overlapping geometry; the TF-activity transforms and learned encoders produce **tighter cell-type clusters**. So "PCA wins" is *probe-specific* — on the intrinsic-structure metric the biological representations come out ahead. Reporting both matters: they answer different questions and here they disagree. (Clustering scores are modest overall — the 15 types include very similar T-cell subtypes — so this qualifies rather than overturns the story.)

Interpretation

The GRN is not useless — as a TF-activity transform it encodes real regulatory signal and gives a genuine low-data/regularization benefit (stronger with CollecTRI). But it is **not more informative than raw-expression PCA**, which captures the same cell-type structure more compactly; and **injecting it into a learned encoder is strictly counterproductive**, adding constraint whose only measurable effect (rewired-equivalent) is regularization. The honest one-liner: *use the GRN as a feature transform, not as a network prior on a deep model, and expect help mainly when data is limited.*

Related work (context)

This matches a clear recent pattern: biology-wired networks rarely beat strong simple baselines on accuracy, and reported gains often reflect capacity/regularization or dimensionality rather than biology. Plain logistic regression matches scBERT on cell typing (Nat Mach Intell 2024); *"one PCA still rules them all"* for perturbation prediction (Nat Methods 2025); biologically-informed decoders (expiMap, Nat Cell Biol 2023; P-NET, Nature 2021) are framed as **interpretability**, not accuracy, wins; and randomly-wired biological nets often match knowledge-primed ones — which is why a shuffled/rewired-graph control is the demanded ablation (Kong et al. 2023). Biologically-informed nets help mainly under small samples / weak signal (2025), consistent with our low-data result. The canonical *successful* use of DoRothEA is exactly ours: decoupler ULM TF-activity (SCENIC/ AUCell lineage, Nat Methods 2017), now often superseded by CollecTRI (NAR 2023). Notably, Arc's "virtual cell" STATE model uses no GRN prior at all. Our donor-held-out design addresses the pseudoreplication/donor-confound pitfall central to between-subjects disease scRNA-seq (Squair et al., Nat Commun 2021).

Limitations

Single primary dataset (COVID as replication); n=500 pseudobulk; RA disease confounded with donor; 5 seeds; one encoder family; probes = logistic/kNN; TF-activity dims differ by network (293 vs 675) so the CollecTRI arm is not dimension-matched to DoRothEA; no formal HPO (by design — see below).

On hyperparameter optimization

Deliberately none. The brief states peak performance is not the goal; all models share architecture/budget *by design*, so tuning one more than another would break the fairness that makes the comparison valid (scIB's benchmark likewise used defaults). Instead we swept prior strength (soft-lambda), ran 5 seeds for variance, used matched-dimension and rewired controls, and swept the bottleneck dimension (32/64/128) — the robustness checks that actually guard the conclusion. The bottleneck sweep confirms the ordering is stable and the hard mask is always worst, so the result is not a z=64 tuning artifact.

Robustness to the early-stopping choice

The autoencoders early-stop on a held-out slice of the *training* donors (never the test donors, so no evaluation leakage — but it does select on a small, noisy val set). We re-ran the full sweep with a **fixed epoch budget on all training donors, no early stopping** (`12_final_fixedbudget.py`, `final_fixedbudget.csv`): the verdict is unchanged — **mean |dF1| = 0.004, max = 0.025**, ordering identical. So the conclusion is not an artifact of early stopping.

Next steps & ideas

Models / architecture - Supervised & transfer embeddings. Train the encoder to predict one label (cell type), then probe a *different* one (disease) — a non-circular transfer test (semi-supervised, à la scANVI); or compare supervised dense-vs-graph-masked to ask whether the prior helps a *classifier* rather than an unsupervised representation. (Same-label supervised eval is circular — excluded.) - **A real GNN.** We covered masked-MLP + soft penalty; a message-passing GNN (GCN/GAT over the DoRothEA graph) is the one learned-prior family untested — it could use higher-order structure. - **Prior as init/regularizer, not hard constraint.** Warm-start a dense encoder from the graph, or add TF-activity prediction as an *auxiliary loss* — softer injections that might keep the benefit without the capacity cost. (Decoder placement already helped; this is the next axis.)

The prior itself — likely the biggest lever - Cell-type-specific / context networks. DoRothEA is a single *global* consensus network, but regulation is context-specific — a global prior may be *why* it underperformed. Use per-cell-type GRNs (SCENIC) or context-matched networks. - **Combine / weight priors.** CollecTRI >= DoRothEA; try ensembling networks, or using DoRothEA's confidence/fractional edge weights instead of binarized +/-1.

Evaluation - Full scIB panel. We added KMeans ARI/NMI; add Leiden + cLISI/kBET/silhouette/isolated-label-F1 + batch metrics for a canonical, metric-robust readout. - **Interpretability payoff.** The literature says biology-wired nets win on *interpretability*, not accuracy — evaluate whether the masked/TF-activity models surface the *correct* driver TFs. - **Proper statistics.** Mixed-effects models accounting for donor structure; more seeds/folds; CIs.

Data & design — breaks the core limitations - A genuinely regulatory readout (perturbation response / measured TF-activity / cell-state transitions) rather than cell-type identity, which linear methods already solve. - **Within-subjects / longitudinal design** (same donor before/during/after) to break the donor–disease confound that caps the disease readout. - **Scale question:** does the prior's low-data benefit vanish as data grows (the STATE / foundation- model regime)? Pretrain-then-finetune to test.

What I deliberately left out (48h scope)

Raw single-cell modeling; STATE/metabolic/pathway priors (out of scope); a full GNN (masked-MLP + soft penalty covered the "learned prior" family); formal HPO (a bottleneck-dim *sensitivity* sweep is included instead). I prioritized a fair, controlled comparison of *how the prior is applied* (constraint vs transform), with corruption, matched-dimension, bottleneck-sensitivity, and second-dataset controls, over broad but shallow coverage.